MLOPS Assignment 2

Tracking a pre-trained sentiment model with ML flow

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Objective: To track the parameters, metrics and artifacts of a pre-defined sentiment-analysis model using ML flow and compare multiple experiment runs.

Technologies and Tools used:

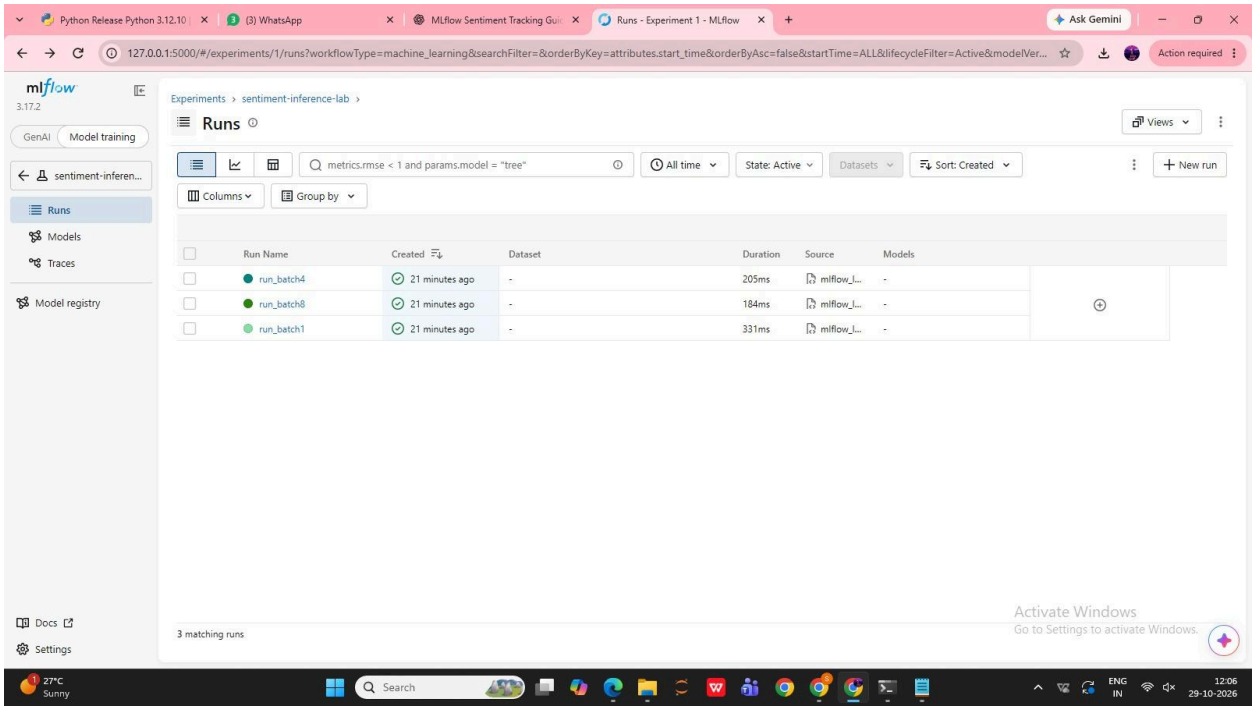
Tools	Purpose
Python 3.12.10	Runs the scripts
MLflow 3.17.2	Tracks and compares experiments
Hugging face transformers	Loads the pre-trained model
PyTorch	Performs sentiment prediction
Pandas	Stores results in CSV

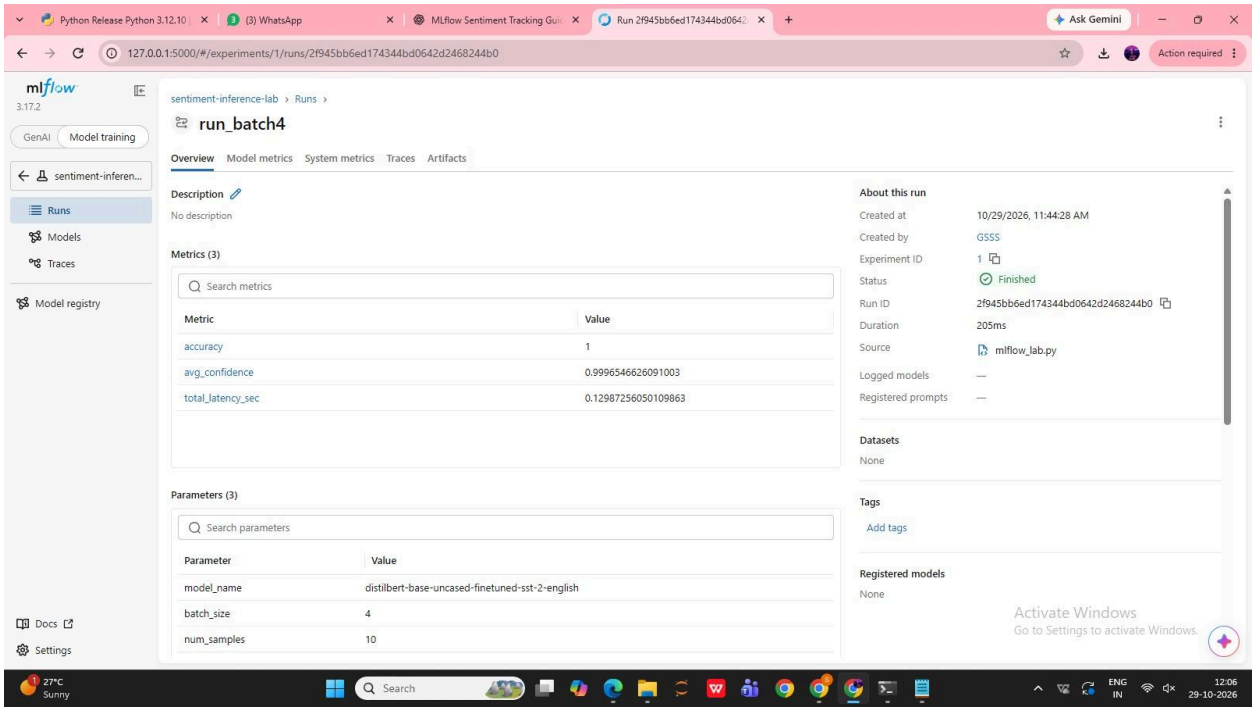
Procedure:

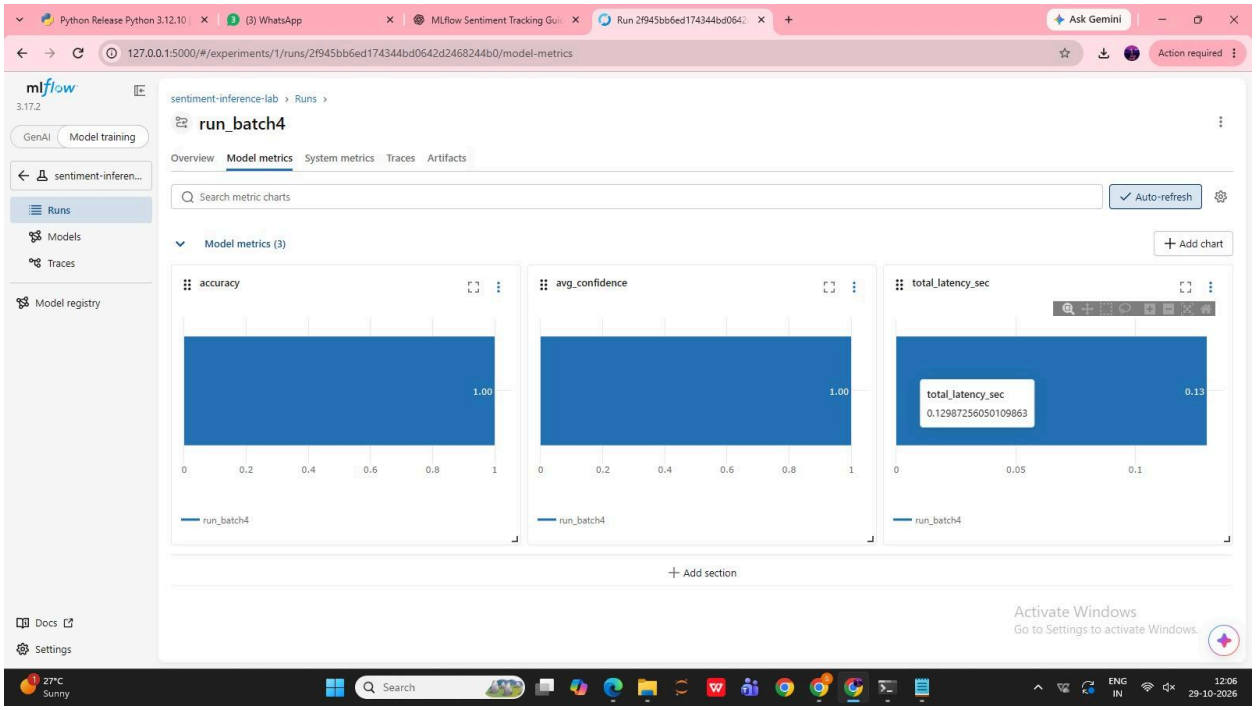
1. Installed the required Python packages and downloaded the pre-trained sentiment model.
2. Executed the model on 10 labelled sentences using batch sizes 1,8,4.
3. Logged the parameters, accuracy, average confidence, latency and predictions.csv artifact using MLflow
4. Open the MLflow UI and compare the three experiment runs.

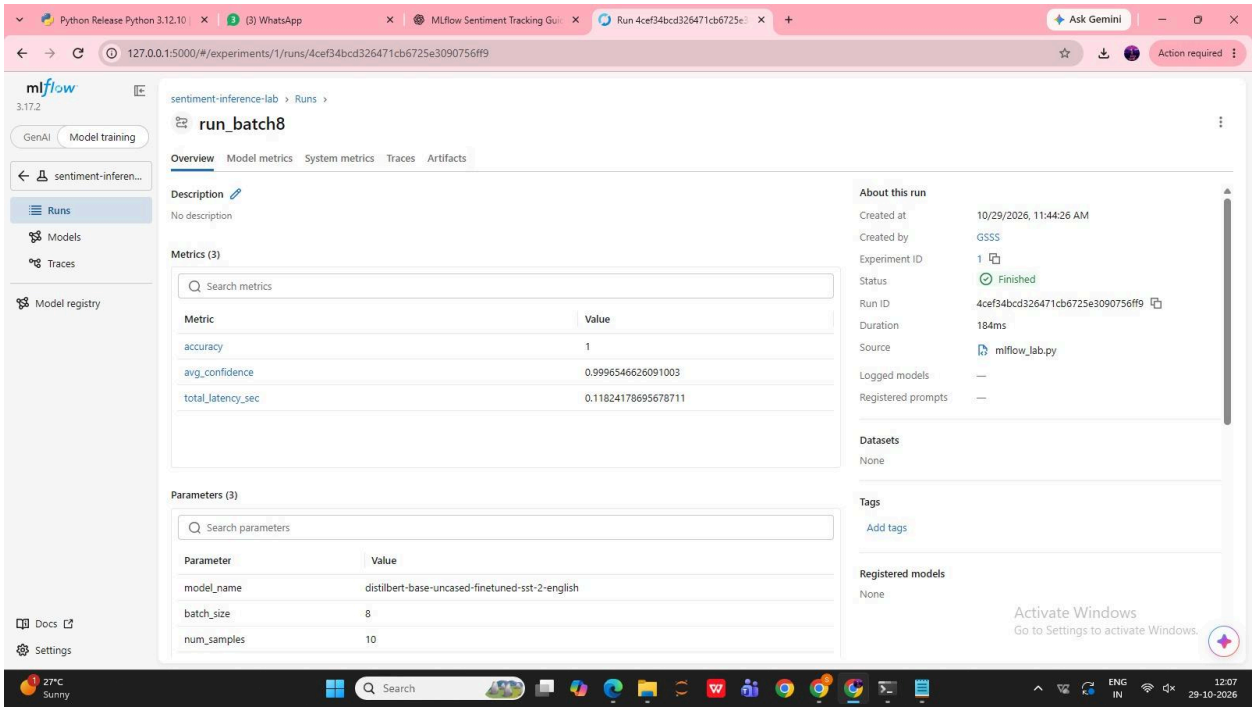
Results:

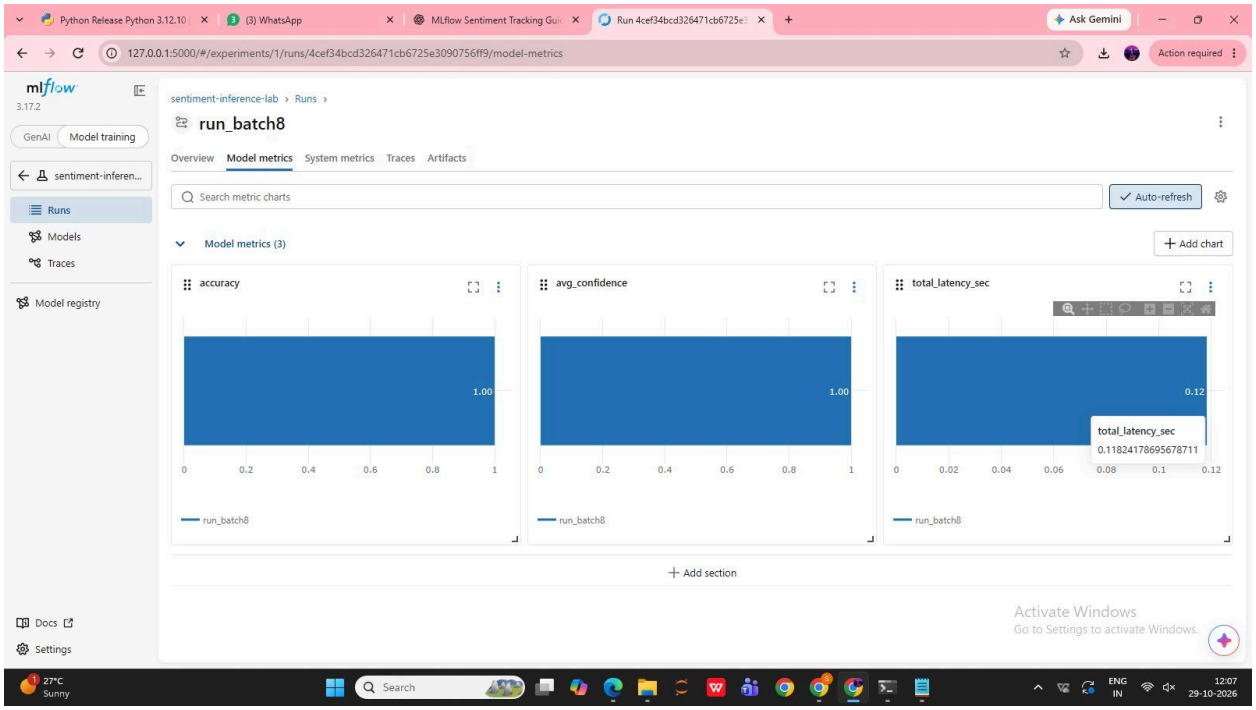
MLflow run table

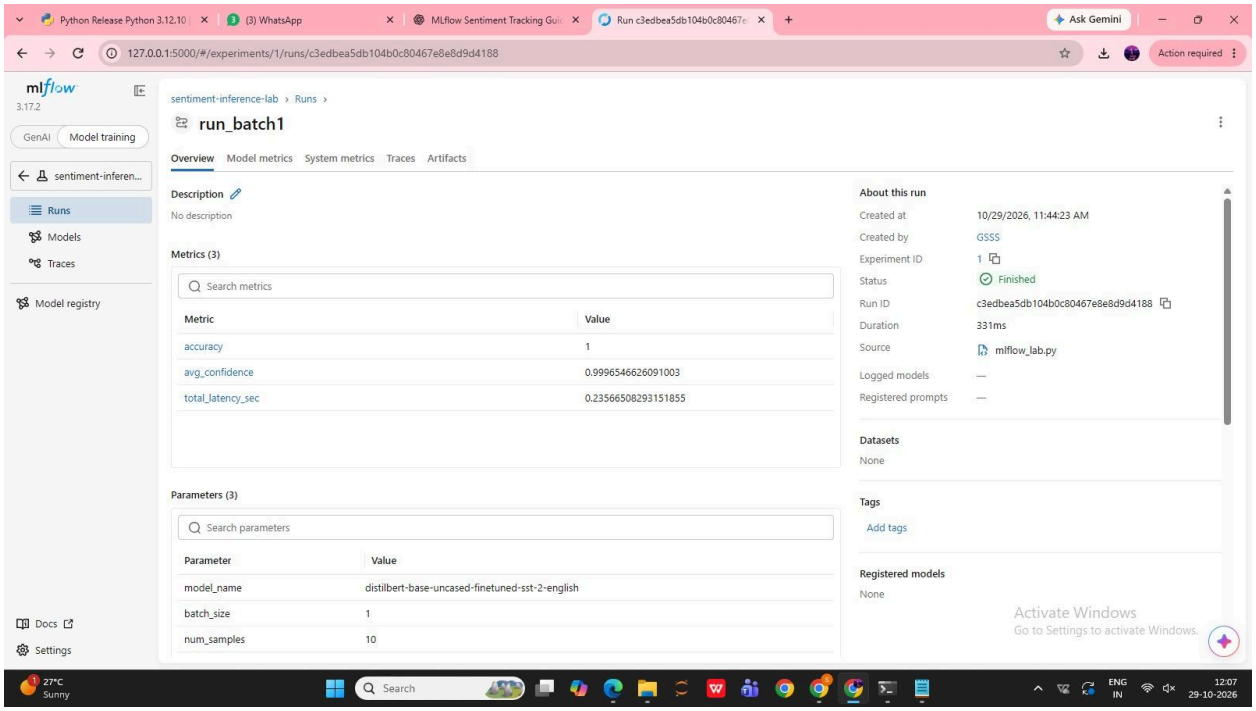

Python Release Python 3.12.10
(3) WhatsApp
MLflow Sentiment Tracking Gui
Runs - Experiment 1 - MLflow
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs?workflowType=machine_learning&searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL&lifecycleFilter=Active&modelVer...
Action required
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
Experiments › sentiment-inference-lab ›
Runs
Views
metrics.rmse < 1 and params.model = "tree"
All time
State: Active
Datasets
Sort: Created
New run
Columns
Group by
Run Name
Created
Dataset
Duration
Source
Models
run_batch4
21 minutes ago
205ms
mlflow_l...
run_batch8
21 minutes ago
184ms
mlflow_l...
run_batch1
21 minutes ago
331ms
mlflow_l...
3 matching runs
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
IN
12:06
29-10-2026

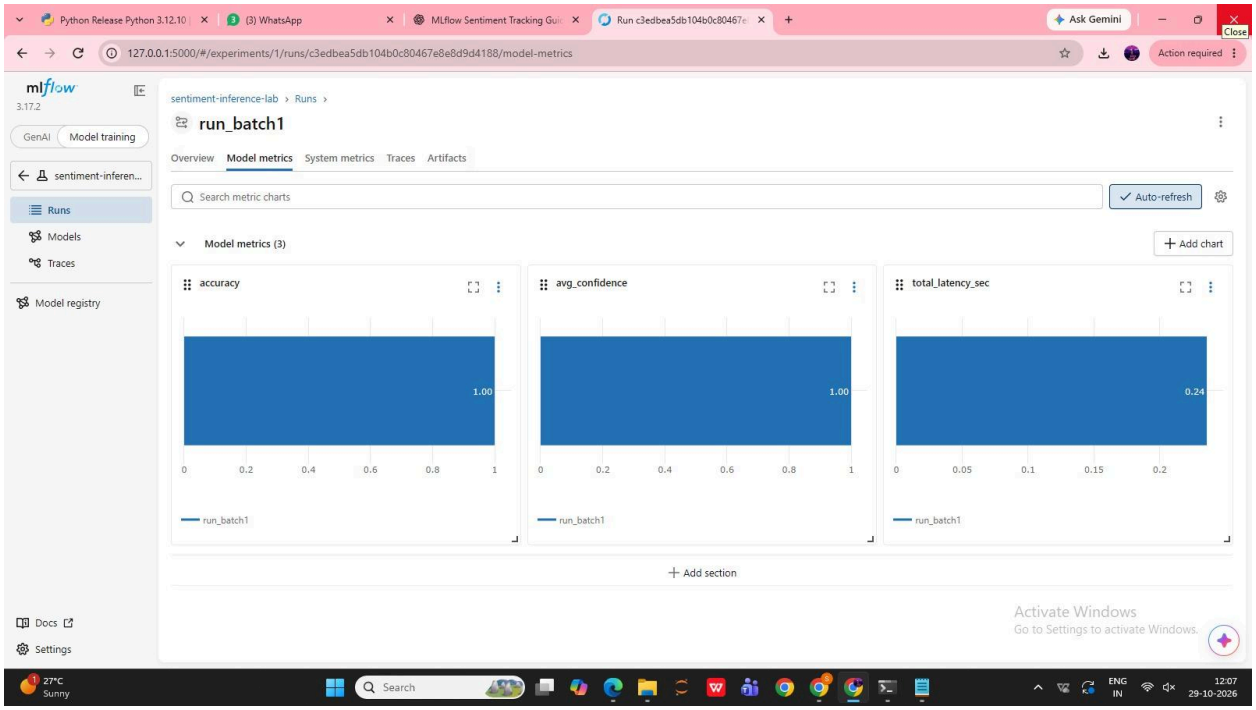

Python Release Python 3.12.10
(3) WhatsApp
MLflow Sentiment Tracking Gui
Run 2f945bb6ed174344bd0642...
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/2f945bb6ed174344bd0642d2468244b0
Action required
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.12987256050109863
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
4
num_samples
10
About this run
Created at
10/29/2026, 11:44:28 AM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
2f945bb6ed174344bd0642d2468244b0
Duration
205ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
IN
12:06
29-10-2026


Python Release Python 3.12.10
(3) WhatsApp
MLflow Sentiment Tracking Gui
Run 2f945bb6ed174344bd0642
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/2f945bb6ed174344bd0642d2468244b0/model-metrics
Action required
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.13
total_latency_sec
0.12987256050109863
0
0.2
0.4
0.6
0.8
1
0.05
0.1
run_batch4
Add section
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
IN
12:06
29-10-2026

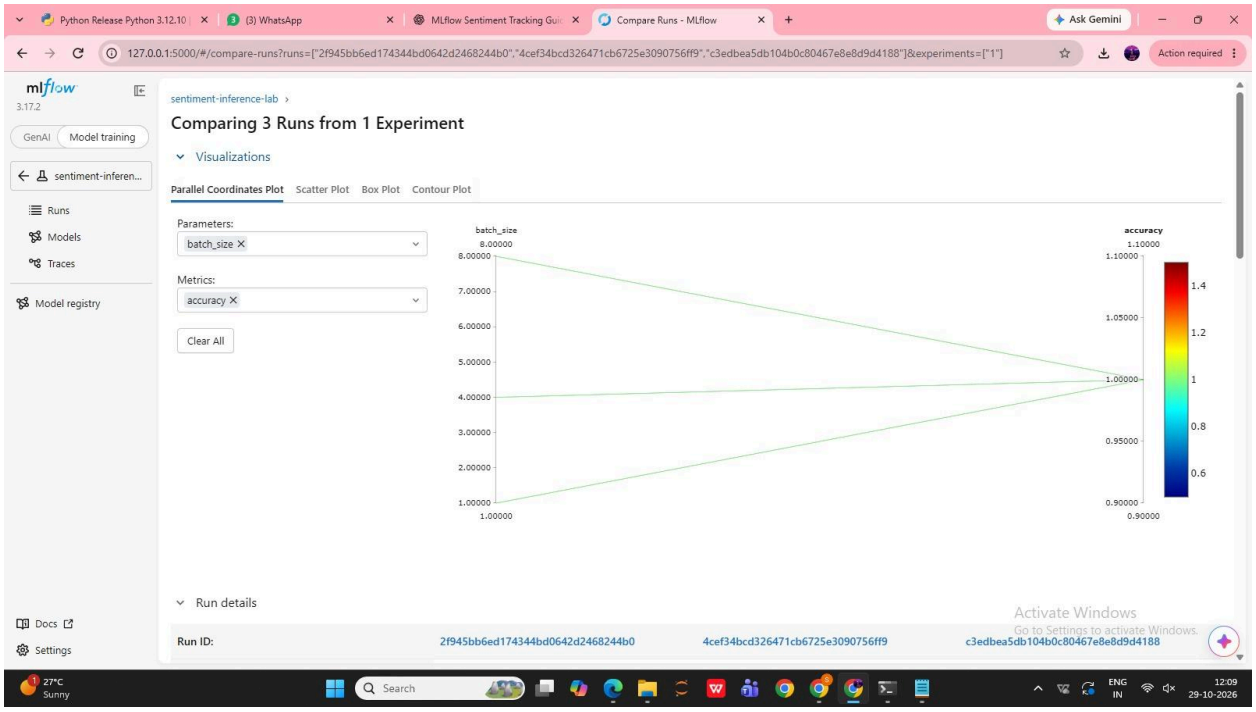

Python Release Python 3.12.10
(3) WhatsApp
MLflow Sentiment Tracking Gui
Run 4cef34bcd326471cb6725e
Ask Gemini
127.0.0.1:5000/#/experiments/1/runs/4cef34bcd326471cb6725e3090756ff9
Action required
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.11824178695678711
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
8
num_samples
10
About this run
Created at
10/29/2026, 11:44:26 AM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
4cef34bcd326471cb6725e3090756ff9
Duration
184ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Activate Windows
Go to Settings to activate Windows.
27°C
Sunny
Search
ENG
IN
12:07
29-10-2026


sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.12
total_latency_sec
0.11824178695678711
run_batch8
Add section
Runs
Models
Traces
Model registry
Docs
Settings

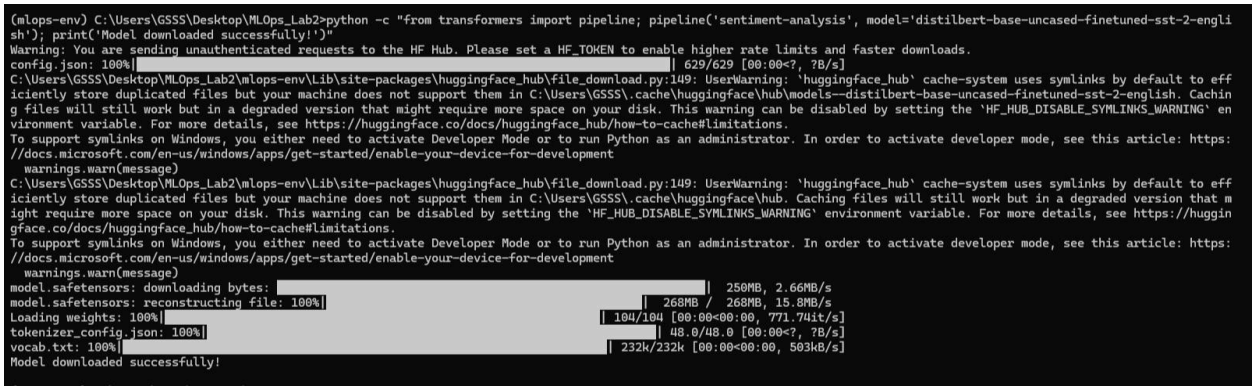

sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.23566508293151855
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
1
num_samples
10
About this run
Created at
10/29/2026, 11:44:23 AM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
c3edbea5db104b0c80467e8e8d9d4188
Duration
331ms
Source
mlflow_lab.py
Logged models
Registered prompts
Datasets
None
Tags
Add tags
Registered models
None
Runs
Models
Traces
Model registry
Docs
Settings

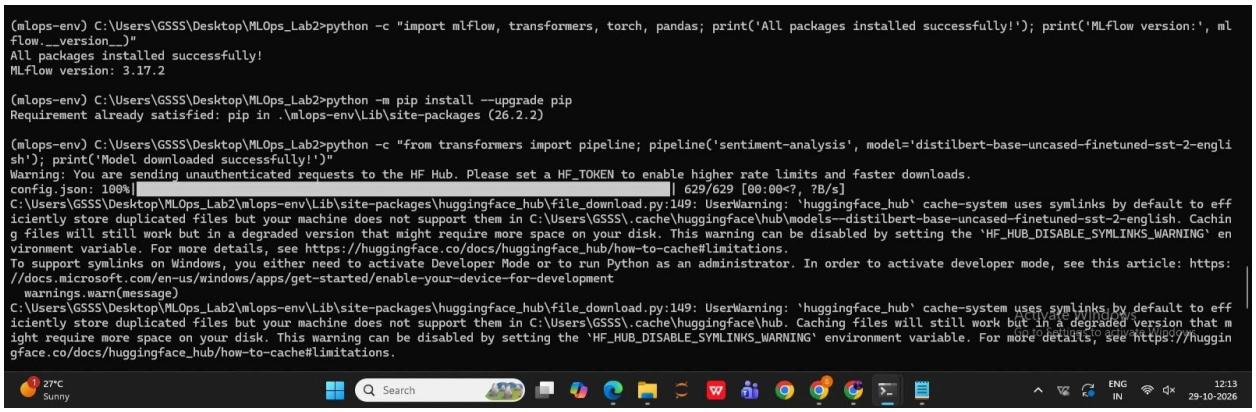


MLflow compare view

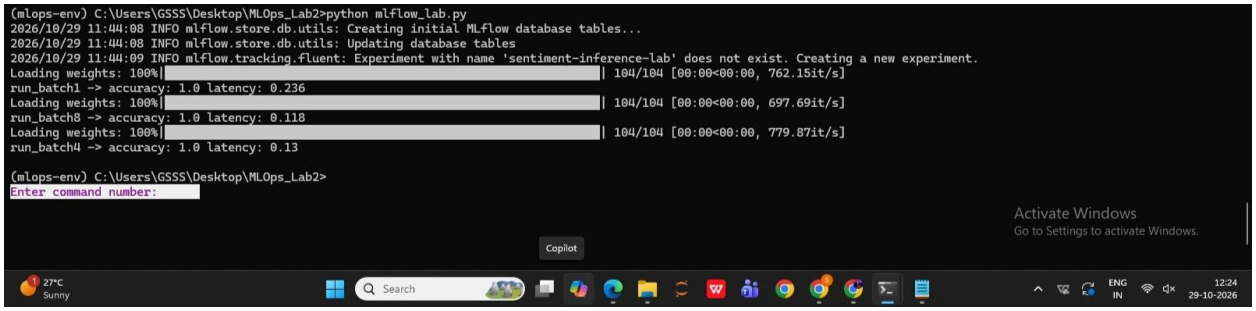


Installations in cmd

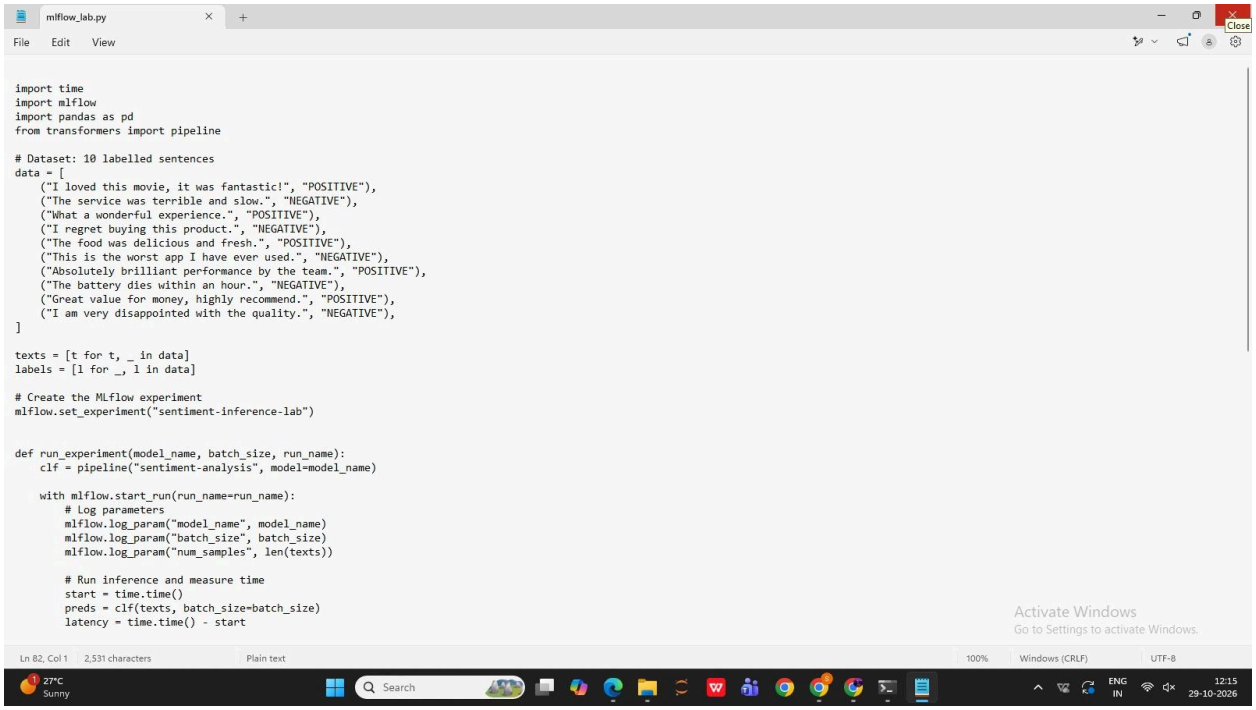




Latency and accuracy



Code:



```
import time
import mlflow
import pandas as pd
from transformers import pipeline

# Dataset: 10 labelled sentences
data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]

texts = [t for t, _ in data]
labels = [l for _, l in data]

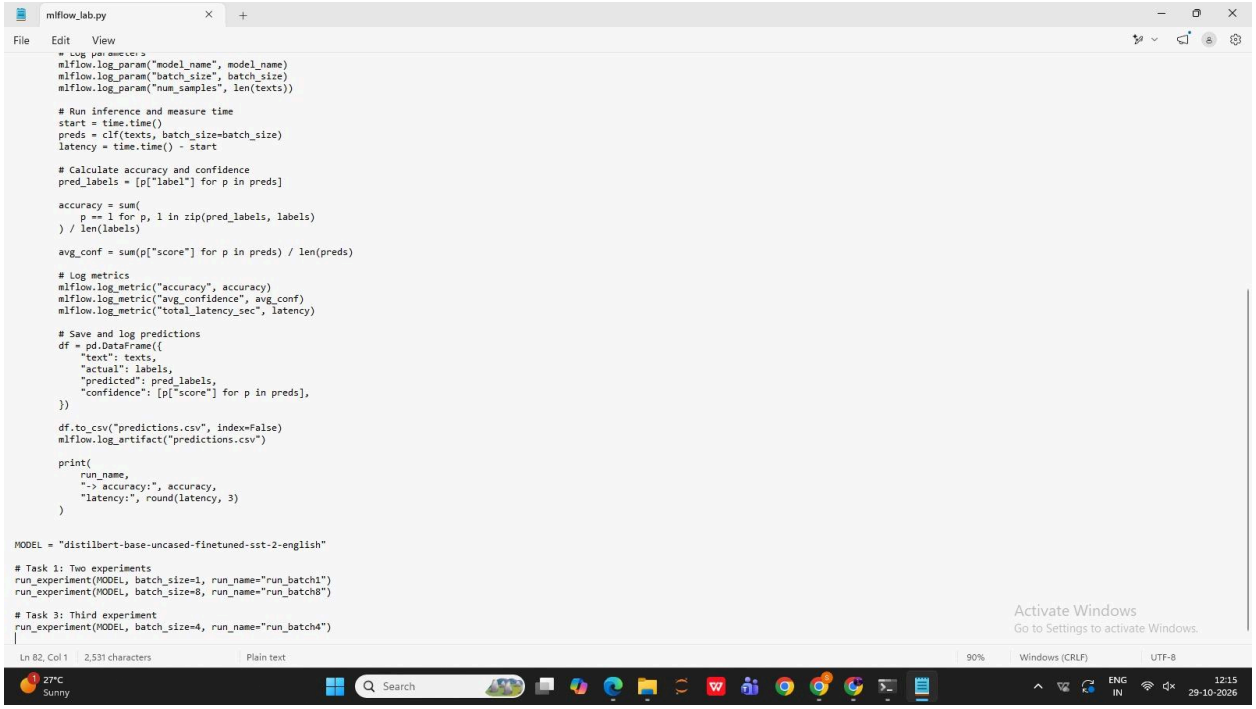
# Create the MLflow experiment
mlflow.set_experiment("sentiment-inference-lab")


def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)

    with mlflow.start_run(run_name=run_name):
        # Log parameters
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        # Run inference and measure time
        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start
```

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```
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        # Run inference and measure time
        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        # Calculate accuracy and confidence
        pred_labels = [p["label"] for p in preds]

        accuracy = sum(
            p == l for p, l in zip(pred_labels, labels)
        ) / len(labels)

        avg_conf = sum(p["score"] for p in preds) / len(preds)

        # Log metrics
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        # Save and log predictions
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })

        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(
            run_name,
            "-> accuracy:", accuracy,
            "latency:", round(latency, 3)
        )


MODEL = "distilbert-base-uncased-finetuned-sst-2-english"

# Task 1: Two experiments
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")

# Task 3: Third experiment
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

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Q&A:

1. Which run had the lowest latency, and why might that be?

run_batch8 had the lowest latency of 0.118 seconds because it processed multiple sentences together.

2. Did accuracy change between runs? Why or why not?

Accuracy remained the same across all three runs, as the same pre-trained sentiment model was used.
All three runs achieved an accuracy of 1.0 (100%)on the 10 test sentences.

3. Name two benefits of using MLflow here instead of printing results in the terminal

- MLflow automatically logs parameters, metrics, and artifacts, making experiment tracking easier.
- The MLflow UI allows easy comparison of multiple runs, unlike printing results only in the terminal.